

VLLM专题（二十二）——生产环境指标

原创

AI专题精讲

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vLLM 提供了许多可用于监控系统健康状况的指标。这些指标通过 vLLM OpenAI 兼容 API 服务器的 `/metrics` 端点公开。

您可以使用 Python 或 Docker 启动服务器：

```
vllm serve unsloth/Llama-3.2-1B-Instruct
```

然后查询该端点以从服务器获取最新的指标：

```
$ curl http://0.0.0.0:8000/metrics

# HELP vllm:iteration_tokens_total Histogram of number of tokens per engine_step.
# TYPE vllm:iteration_tokens_total histogram
vllm:iteration_tokens_total_sum{model_name="unsloth/Llama-3.2-1B-Instruct"} 0.0
vllm:iteration_tokens_total_bucket{le="1.0",model_name="unsloth/Llama-3.2-1B-Instruct"} 3.0
vllm:iteration_tokens_total_bucket{le="8.0",model_name="unsloth/Llama-3.2-1B-Instruct"} 3.0
vllm:iteration_tokens_total_bucket{le="16.0",model_name="unsloth/Llama-3.2-1B-Instruct"} 3.0
vllm:iteration_tokens_total_bucket{le="32.0",model_name="unsloth/Llama-3.2-1B-Instruct"} 3.0
vllm:iteration_tokens_total_bucket{le="64.0",model_name="unsloth/Llama-3.2-1B-Instruct"} 3.0
vllm:iteration_tokens_total_bucket{le="128.0",model_name="unsloth/Llama-3.2-1B-Instruct"} 3.0
vllm:iteration_tokens_total_bucket{le="256.0",model_name="unsloth/Llama-3.2-1B-Instruct"} 3.0
vllm:iteration_tokens_total_bucket{le="512.0",model_name="unsloth/Llama-3.2-1B-Instruct"} 3.0
...
```

以下是公开的指标：

```
class Metrics:
    """
    vLLM uses a multiprocessing-based frontend for the OpenAI server.
    This means that we need to run prometheus_client in multiprocessing mode
    See https://prometheus.github.io/client_python/multiprocess/ for more
    details on limitations.
    """

    labelname_finish_reason = "finished_reason"
    labelname_waiting_lora_adapters = "waiting_lora_adapters"
    labelname_running_lora_adapters = "running_lora_adapters"
    labelname_max_lora = "max_lora"
```

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```

_gauge_cls = prometheus_client.Gauge
_counter_cls = prometheus_client.Counter
_histogram_cls = prometheus_client.Histogram

def __init__(self, labelnames: List[str], vllm_config: VllmConfig):
    # Unregister any existing vLLM collectors (for CI/CD)
    self.unregister_vllm_metrics()

    max_model_len = vllm_config.model_config.max_model_len

    # System stats
    # Scheduler State
    self.gauge_scheduler_running = self._gauge_cls(
        name="vllm:num_requests_running",
        documentation="Number of requests currently running on GPU.",
        labelnames=labelnames,
        multiprocess_mode="sum")
    self.gauge_scheduler_waiting = self._gauge_cls(
        name="vllm:num_requests_waiting",
        documentation="Number of requests waiting to be processed.",
        labelnames=labelnames,
        multiprocess_mode="sum")
    self.gauge_lora_info = self._gauge_cls(
        name="vllm:lora_requests_info",
        documentation="Running stats on lora requests.",
        labelnames=[
            self.labelname_running_lora_adapters,
            self.labelname_max_lora,
            self.labelname_waiting_lora_adapters,
        ],
        multiprocess_mode="livemostrecent",
    )

    # Deprecated in 0.8 - KV cache offloading is not used in V1
    # TODO: in 0.9, only enable if show_hidden_metrics=True
    self.gauge_scheduler_swapped = self._gauge_cls(
        name="vllm:num_requests_swapped",
        documentation=(
            "Number of requests swapped to CPU. "
            "DEPRECATED: KV cache offloading is not used in V1"),
        labelnames=labelnames,
        multiprocess_mode="sum")

    # KV Cache Usage in %
    self.gauge_gpu_cache_usage = self._gauge_cls(
        name="vllm:gpu_cache_usage_perc",
        documentation="GPU KV-cache usage. 1 means 100 percent usage.",

```

```

        labelnames=labelnames,
        multiprocess_mode="sum")

# Deprecated in 0.8 - KV cache offloading is not used in V1
# TODO: in 0.9, only enable if show_hidden_metrics=True
self.gauge_cpu_cache_usage = self._gauge_cls(
    name="vllm:cpu_cache_usage_perc",
    documentation=(
        "CPU KV-cache usage. 1 means 100 percent usage. "
        "DEPRECATED: KV cache offloading is not used in V1"),
    labelnames=labelnames,
    multiprocess_mode="sum")

# Deprecated in 0.8 - KV cache offloading is not used in V1
# TODO: in 0.9, only enable if show_hidden_metrics=True
self.gauge_cpu_prefix_cache_hit_rate = self._gauge_cls(
    name="vllm:cpu_prefix_cache_hit_rate",
    documentation=(
        "CPU prefix cache block hit rate. "
        "DEPRECATED: KV cache offloading is not used in V1"),
    labelnames=labelnames,
    multiprocess_mode="sum")

# Deprecated in 0.8 - replaced by queries+hits counters in V1
# TODO: in 0.9, only enable if show_hidden_metrics=True
self.gauge_gpu_prefix_cache_hit_rate = self._gauge_cls(
    name="vllm:gpu_prefix_cache_hit_rate",
    documentation=(
        "GPU prefix cache block hit rate. "
        "DEPRECATED: use vllm:gpu_prefix_cache_queries and "
        "vllm:gpu_prefix_cache_queries in V1"),
    labelnames=labelnames,
    multiprocess_mode="sum")

# Iteration stats
self.counter_num_preemption = self._counter_cls(
    name="vllm:num_preemptions_total",
    documentation="Cumulative number of preemption from the engine.",
    labelnames=labelnames)
self.counter_prompt_tokens = self._counter_cls(
    name="vllm:prompt_tokens_total",
    documentation="Number of prefill tokens processed.",
    labelnames=labelnames)
self.counter_generation_tokens = self._counter_cls(
    name="vllm:generation_tokens_total",
    documentation="Number of generation tokens processed.",
    labelnames=labelnames)
buckets = [1, 8, 16, 32, 64, 128, 256, 512, 1024, 2048, 4096, 8096]
if not vllm_config.model_config.enforce_eager:

```

```

        buckets = vllm_config.compilation_config.\
            cudagraph_capture_sizes.copy()
        buckets.sort()
    self.histogram_iteration_tokens = self._histogram_cls(
        name="vllm:iteration_tokens_total",
        documentation="Histogram of number of tokens per engine_step.",
        labelnames=labelnames,
        buckets=buckets)
    self.histogram_time_to_first_token = self._histogram_cls(
        name="vllm:time_to_first_token_seconds",
        documentation="Histogram of time to first token in seconds.",
        labelnames=labelnames,
        buckets=[
            0.001, 0.005, 0.01, 0.02, 0.04, 0.06, 0.08, 0.1, 0.25, 0.5,
            0.75, 1.0, 2.5, 5.0, 7.5, 10.0
        ])
    self.histogram_time_per_output_token = self._histogram_cls(
        name="vllm:time_per_output_token_seconds",
        documentation="Histogram of time per output token in seconds.",
        labelnames=labelnames,
        buckets=[
            0.01, 0.025, 0.05, 0.075, 0.1, 0.15, 0.2, 0.3, 0.4, 0.5, 0.75,
            1.0, 2.5
        ])

# Request stats
# Latency
request_latency_buckets = [
    0.3, 0.5, 0.8, 1.0, 1.5, 2.0, 2.5, 5.0, 10.0, 15.0, 20.0, 30.0,
    40.0, 50.0, 60.0
]
self.histogram_e2e_time_request = self._histogram_cls(
    name="vllm:e2e_request_latency_seconds",
    documentation="Histogram of end to end request latency in seconds.",
    labelnames=labelnames,
    buckets=request_latency_buckets)
self.histogram_queue_time_request = self._histogram_cls(
    name="vllm:request_queue_time_seconds",
    documentation="Histogram of time spent in WAITING phase for request.",
    labelnames=labelnames,
    buckets=request_latency_buckets)
self.histogram_inference_time_request = self._histogram_cls(
    name="vllm:request_inference_time_seconds",
    documentation="Histogram of time spent in RUNNING phase for request.",
    labelnames=labelnames,

```

```

        buckets=request_latency_buckets)
self.histogram_prefill_time_request = self._histogram_cls(
    name="vllm:request_prefill_time_seconds",
    documentation=
        "Histogram of time spent in PREFILL phase for request.",
    labelnames=labelnames,
    buckets=request_latency_buckets)
self.histogram_decode_time_request = self._histogram_cls(
    name="vllm:request_decode_time_seconds",
    documentation=
        "Histogram of time spent in DECODE phase for request.",
    labelnames=labelnames,
    buckets=request_latency_buckets)
# Deprecated in 0.8 - duplicates vllm:request_queue_time_seconds:
# TODO: in 0.9, only enable if show_hidden_metrics=True
self.histogram_time_in_queue_request = self._histogram_cls(
    name="vllm:time_in_queue_requests",
    documentation=(
        "Histogram of time the request spent in the queue in seconds. "
        "DEPRECATED: use vllm:request_queue_time_seconds instead."),
    labelnames=labelnames,
    buckets=request_latency_buckets)

# Deprecated in 0.8 - use prefill/decode/inference time metrics
# TODO: in 0.9, only enable if show_hidden_metrics=True
self.histogram_model_forward_time_request = self._histogram_cls(
    name="vllm:model_forward_time_milliseconds",
    documentation=(
        "Histogram of time spent in the model forward pass in ms. "
        "DEPRECATED: use prefill/decode/inference time metrics instead."
    ),
    labelnames=labelnames,
    buckets=build_1_2_3_5_8_buckets(3000))
self.histogram_model_execute_time_request = self._histogram_cls(
    name="vllm:model_execute_time_milliseconds",
    documentation=(
        "Histogram of time spent in the model execute function in ms."
        "DEPRECATED: use prefill/decode/inference time metrics instead."
    ),
    labelnames=labelnames,
    buckets=build_1_2_3_5_8_buckets(3000))

# Metadata
self.histogram_num_prompt_tokens_request = self._histogram_cls(
    name="vllm:request_prompt_tokens",
    documentation="Number of prefill tokens processed.",
    labelnames=labelnames,
    buckets=build_1_2_5_buckets(max_model_len)

```

```

        buckets=build_1_2_5_buckets(max_model_len),
    )
    self.histogram_num_generation_tokens_request = \
        self._histogram_cls(
            name="vllm:request_generation_tokens",
            documentation="Number of generation tokens processed.",
            labelnames=labelnames,
            buckets=build_1_2_5_buckets(max_model_len),
        )
    self.histogram_max_num_generation_tokens_request = self._histogram_cls(
        name="vllm:request_max_num_generation_tokens",
        documentation=
            "Histogram of maximum number of requested generation tokens.",
        labelnames=labelnames,
        buckets=build_1_2_5_buckets(max_model_len))
    self.histogram_n_request = self._histogram_cls(
        name="vllm:request_params_n",
        documentation="Histogram of the n request parameter.",
        labelnames=labelnames,
        buckets=[1, 2, 5, 10, 20],
    )
    self.histogram_max_tokens_request = self._histogram_cls(
        name="vllm:request_params_max_tokens",
        documentation="Histogram of the max_tokens request parameter.",
        labelnames=labelnames,
        buckets=build_1_2_5_buckets(max_model_len),
    )
    self.counter_request_success = self._counter_cls(
        name="vllm:request_success_total",
        documentation="Count of successfully processed requests.",
        labelnames=labelnames + [Metrics.labelname_finish_reason])

    # Speculative decoding stats
    self.gauge_spec_decode_draft_acceptance_rate = self._gauge_cls(
        name="vllm:spec_decode_draft_acceptance_rate",
        documentation="Speulative token acceptance rate.",
        labelnames=labelnames,
        multiprocess_mode="sum")
    self.gauge_spec_decode_efficiency = self._gauge_cls(
        name="vllm:spec_decode_efficiency",
        documentation="Speculative decoding system efficiency.",
        labelnames=labelnames,
        multiprocess_mode="sum")
    self.counter_spec_decode_num_accepted_tokens = (self._counter_cls(
        name="vllm:spec_decode_num_accepted_tokens_total",
        documentation="Number of accepted tokens.",
        labelnames=labelnames))
    self.counter_spec_decode_num_draft_tokens = self._counter_cls(

```

```
name="vllm:spec_decode_num_draft_tokens_total",
documentation="Number of draft tokens.",
labelnames=labelnames)
self.counter_spec_decode_num_emitted_tokens = (self._counter_cls(
name="vllm:spec_decode_num_emitted_tokens_total",
documentation="Number of emitted tokens.",
labelnames=labelnames))
```

以下指标已被弃用，并将在未来版本中移除：

- **vllm:num_requests_swapped**、**vllm:cpu_cache_usage_perc** 和 **vllm:cpu_prefix_cache_hit_rate**：因为在 V1 中未使用 KV 缓存卸载功能。
- **vllm:gpu_prefix_cache_hit_rate**：在 V1 中被 `queries+hits` 计数器取代。
- **vllm:time_in_queue_requests**：因为它与 `vllm:request_queue_time_seconds` 重复。
- **vllm:model_forward_time_milliseconds** 和 **vllm:model_execute_time_milliseconds**：因为应改用预填充/解码/推理时间指标。

注意：当指标在版本 X.Y 中被弃用时，它们会在版本 X.Y+1 中被隐藏，但可以使用 `--show-hidden-metrics-for-version=X.Y` 参数重新启用，然后在版本 X.Y+2 中完全移除。



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